Third-party notices

RSAS is released under the MIT License (see `LICENSE.md`). It bundles and depends on third-party tools that carry their own licenses. RSAS invokes the bundled binaries as **separate executables** (via subprocess); they are aggregated with RSAS, not linked into it, so each remains under its own license and RSAS itself stays MIT.

Full license texts and per-component notices are in the `LICENSES/` directory (`GPL-3.0.txt` and `THIRD-PARTY-NOTICE.txt`), and ship inside the packaged app's DMG.

Bundled binaries (in `bin/`)

RNArobo

- **Version:** 2.1.0 (September 2013)
- **License:** GNU General Public License v3.0 (`LICENSES/GPL-3.0.txt`). GPL-3.0 permits commercial and non-commercial use, modification, and redistribution, provided the license and copyright notices are preserved and the corresponding source remains available. RSAS redistributes it unmodified.
- **Source:** https://github.com/rampasek/RNArobo; release archives and the Linux executable: https://compbio.fmph.uniba.sk/software.php (Computational Biology, Comenius University in Bratislava)
- **Use in RSAS:** structural motif search (the RNArobo Search page).
- **Citation:** Rampášek, L., Jimenez, R.M., Lupták, A., Vinař, T., Brejová, B. (2016). RNA motif search with data-driven element ordering. *BMC Bioinformatics* 17:216.

Knotty

- **License:** GNU General Public License (stated in the upstream README; copyright University of Alberta and the MultiRNAFold/SimFold authors). Redistributed unmodified.
- **Source:** https://github.com/HosnaJabbari/Knotty
- **Use in RSAS:** pseudoknot prediction (the Pseudoknot Finder page), using the DP09 energy model.
- **Citation:** Jabbari, H., Wark, I., Montemagno, C., Will, S. (2018). Knotty: efficient and accurate prediction of complex RNA pseudoknot structures. *Bioinformatics* 34(22):3849-3856.

SimFold energy parameters (in `bin/macos/simfold/params/`)

- Knotty reads its thermodynamic parameters from SimFold / MultiRNAFold (Turner and Andronescu/Mathews parameter sets), which are part of MultiRNAFold and distributed under the same GPL terms as Knotty.
- **Source:** http://www.rnasoft.ca/download.html

Bundled / runtime dependencies

ViennaRNA

- RSAS performs all folding through ViennaRNA's Python bindings. The packaged macOS application bundles ViennaRNA; when running from source it is installed separately by the user.
- **Citation:** Lorenz, R. et al. (2011). ViennaRNA Package 2.0. *Algorithms for Molecular Biology* 6:26.
- **Project / license:** https://www.tbi.univie.ac.at/RNA/

Other libraries

- Biopython, NumPy, openpyxl, CustomTkinter and Pillow are bundled in the packaged application, each under its own permissive open-source license.